

Continuum Limit: Derivation and Applications in Mathematical Models

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Summary

Motivation

In the real world, many systems involve a **huge number** of interacting units:

- Power grids with millions of nodes;
- Neurons, oscillators, or agents in networks;
- Social/biological populations with many individuals.

Discrete models are natural but often intractable for large systems.

Motivation (why continuum limit?)

Example: Discrete power network (Swing equations)

- θ_i : phase of the i -th electronic component in a power network;
- α_i : damping;
- P_i : power injection;
- K_{ij} : coupling coefficient between nodes i and j .

Motivation (why continuum limit?)

Example: Discrete power network (Swing equations)

Node $i = 1, \dots, N$

$$\ddot{\theta}_i(t) + \alpha_i \dot{\theta}_i(t) = P_i + \frac{1}{N} \sum_{j=1}^N K_{ij} \sin(\theta_j(t) - \theta_i(t)). \quad (1.1)$$

Motivation (why continuum limit?)

For a given dynamical system, a central question is whether **equilibria exist** and whether they are **stable**.

ODE network model (1.1):

- Linearize the $2N$ -dimensional ODE system at an equilibrium.
- Stability requires analyzing a $(2N) \times (2N)$ **Jacobian matrix**.
- For large N , this becomes hard.

Motivation (why continuum limit?)

Continuum limit (integro-PDE)

Position $x \in [0, 1]$ for N large

$$\begin{aligned} \partial_{tt}\theta(t, x) = & -\alpha(x)\partial_t\theta(t, x) + P(x) \\ & + \int_0^1 K(x, y) \sin(\theta(t, y) - \theta(t, x)) \, dy. \end{aligned} \tag{1.2}$$

- Nonlocal coupling via kernel $K(x, y)$
- Relatively easy for stability/bifurcation analysis
- Scalable macroscopic description

Motivation (why continuum limit?)

Continuum integro-PDE model (1.2):

- Linearize the integro-differential equation around a homogeneous equilibrium.
- Stability reduces to analyzing a 2×2 **matrix**, depending on wavenumber or kernel eigenvalues.
- This simplification often leads to **explicit analytic conditions**.

Motivation (why continuum limit?)

Another important question is:

Can the solution of the continuum model approximate the discrete dynamics?

Key considerations:

- **Convergence:** as $N \rightarrow \infty$, differences between discrete and continuum solutions can be estimated.
- **Numerical evidence:** PDE simulations often capture the qualitative patterns of the large- N ODE system.

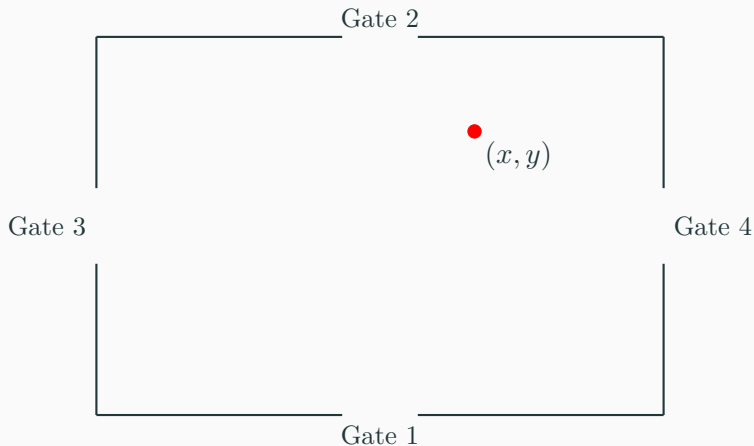
Random Walk with Four Gates

Example: Random Walk with Four Gates

- A stadium is modeled as a bounded rectangular domain $\Omega \subset \mathbb{R}^2$ with walls except for four gates $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4 \subset \partial\Omega$.
- A walker starts at $(x, y) \in \Omega$ and performs a symmetric random walk.
- Let $P(x, y)$ denote the probability that the walker **eventually exits the stadium** through any gate.

What is $P(x, y)$?

Random Walk with Four Gates



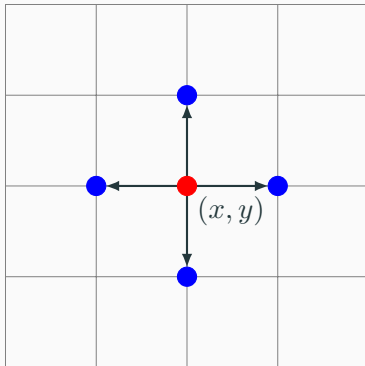
Random Walk with Four Gates

- We discretize the rectangular region into a square lattice with mesh size h .
- The walker's position is described by lattice points (x, y) .
- At each step, the walker moves to one of its four neighbors with equal probability:

$$(x + h, y), \quad (x - h, y), \quad (x, y + h), \quad (x, y - h).$$

- The walker will stop if reaches the wall or leave the stadium.
- This defines a **symmetric random walk** on the grid.

Random Walk with Four Gates



$$P(x, y) = \frac{1}{4} \left[P(x + h, y) + P(x - h, y) \right. \\ \left. + P(x, y + h) + P(x, y - h) \right]. \quad (2.1)$$

Random Walk with Four Gates

Assume $h \ll \text{diam}(\Omega)$,

$$\Delta P(x, y) \approx \frac{1}{h^2} \left[P(x+h, y) + P(x-h, y) + P(x, y+h) + P(x, y-h) - 4P(x, y) \right]. \quad (2.2)$$

Then (2.1) becomes

$$\Delta P(x, y) = \frac{\partial^2 P(x, y)}{\partial x^2} + \frac{\partial^2 P(x, y)}{\partial y^2} = 0. \quad (2.3)$$

To fully determine the solution $P(x, y)$, we must specify **boundary conditions** on $\partial\Omega$. We have:

$$P(x, y) = \begin{cases} 1 & \text{if } (x, y) \in \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4, \\ 0 & \text{if } (x, y) \notin \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4. \end{cases} \quad (2.4)$$

This function is also denoted as χ_Γ , where $\Gamma = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4$.

Final PDE problem:

$$\begin{cases} \Delta P(x, y) = 0 & \text{if } (x, y) \in \Omega, \\ P(x, y) = \chi_{\Gamma} & \text{if } (x, y) \in \partial\Omega. \end{cases} \quad (2.5)$$

Directed Motion

- In **random motion**, individuals move to neighboring colonies with equal probability.
- In **directed motion**, individuals prefer places with higher **attractiveness**.

The migration probability becomes **biased**, for instance:

$$\mathbb{P}(i \rightarrow j) = \frac{f(w, A_j)}{f(w, A_i) + \sum_{k \in N(i)} f(w, A_k)}, \quad (3.1)$$

where $N(i)$ represents the neighbour of i , f is a weight function, and w controls the strength of bias.

Directed Motion

Start with a reaction system:

$$\frac{du_i}{dt} = g(u_i). \quad (3.2)$$

To include spatial effects, consider communities exchanging individuals:

$$\begin{aligned} u_i(t + \Delta t) = & u_i(t) + g(u_i)\Delta t + \sum_{j \in N(i)} \mathbb{P}(j \rightarrow i)u_j(t) \\ & - \sum_{j \in N(i)} \mathbb{P}(i \rightarrow j)u_i(t). \end{aligned} \quad (3.3)$$

The last two terms describe migration (inflow and outflow).

Directed Motion

Consider regular lattice in 2D (or n D), with spacing h . In the denominator,

$$\sum_{k \in N(i)} f(w, A_k) = h^2 \Delta f(w, A_i) + 4f(w, A_i) + o(h^2).$$

Then

$$\begin{aligned} \mathbb{P}(i \rightarrow j) &= \frac{f(w, A_j)}{5f(w, A_i) + h^2 \Delta f(w, A_i) + o(h^2)} \\ &= f(w, A_j) \left[\frac{1}{5f(w, A_i)} - \frac{h^2 \Delta f(w, A_i)}{25f^2(w, A_i)} \right] + O(h^4). \end{aligned}$$

Here we use $1/(a+x) = 1/a - x/a^2 + o(x)$.

Then (3.3) becomes

$$\begin{aligned} u_i(t + \Delta t) - u_i(t) - g(u_i)\Delta t = & \frac{h^2}{5} \left[f(w, A_i) \Delta \left(\frac{u_i}{f(w, A_i)} \right) \right. \\ & \left. - \frac{u_i}{f(w, A_i)} \Delta f(w, A_i) \right] + O(h^4) \end{aligned}$$

Let

$$X = \frac{u_i}{f(w, A_i)}, \quad Y = f(w, A_i),$$

we use the fact that

$$\Delta(XY) = Y\Delta X + X\Delta Y + 2\nabla X \cdot \nabla Y.$$

$$\begin{aligned} u_i(t + \Delta t) - u_i(t) &= \frac{h^2}{5} \left[\Delta u_i - 2 \nabla f(w, A_i) \cdot \nabla \left(\frac{u_i}{f(w, A_i)} \right) \right. \\ &\quad \left. - \frac{2u_i}{f(w, A_i)} \Delta f(w, A_i) \right] + g(u_i) \Delta t + O(h^4) \\ &= \frac{h^2}{5} \nabla \cdot \left[\nabla u_i - \frac{2u_i}{f(w, A_i)} \nabla f(w, A_i) \right] \\ &\quad + g(u_i) \Delta t + O(h^4) \\ &= \frac{h^2}{5} \nabla \cdot \left[\nabla u_i - 2u_i \nabla \ln f(w, A_i) \right] \\ &\quad + g(u_i) \Delta t + O(h^4). \end{aligned}$$

Assume $h^2/\Delta t = \text{constant}$. Also, write $\chi_w(u) = \ln f(w, A)$.
Then divide by Δt and take the limit $\Delta t \rightarrow 0^+$,

$$\frac{\partial u(t, \mathbf{x})}{\partial t} = D \nabla \cdot [\nabla u - 2u \nabla \chi_w(u)] + g(u). \quad (3.4)$$

Example (public goods dilemma): a situation where individuals must decide whether to contribute to a shared resource that benefits everyone but involves a personal cost (e.g., time, effort, or health).

Directed Motion

A typical ODE model for the public goods dilemma:

$$\begin{cases} \frac{du_i}{dt} = u_i [r_u \phi_i - c - \gamma_i(u_i + v_i) - \mu_u], \\ \frac{dv_i}{dt} = v_i [r_v \phi_i - \gamma_i(u_i + v_i) - \mu_v], \\ \frac{d\phi_i}{dt} = cu_i - [\kappa(u_i + v_i) + \delta] \phi_i. \end{cases}$$

Here

- u_i : number of the cooperators;
- v_i : number of the defectors;
- ϕ_i : total amount of the public goods.

$$\left\{ \begin{array}{l} \frac{\partial u(t, \mathbf{x})}{\partial t} = D_u \nabla \cdot [\nabla u - 2w_u u \nabla \phi] \\ \quad + u [r_u \phi - c - \gamma(u + v) - \mu_u], \\ \frac{\partial v(t, \mathbf{x})}{\partial t} = D_v \nabla \cdot [\nabla v - 2w_v v \nabla \phi] \\ \quad + v [r_v \phi - \gamma(u + v) - \mu_v], \\ \frac{\partial \phi(t, \mathbf{x})}{\partial t} = D_\phi \nabla^2 \phi + cu - [\kappa(u + v) + \delta] \phi. \end{array} \right.$$

Nonlocal Directed Motion

So far:

- pure random walk (local) $\Rightarrow$

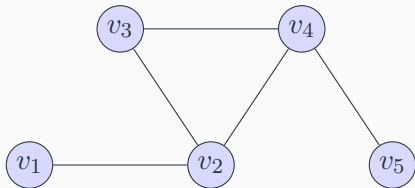
$$\Delta u = 0.$$

- local directed motion $\Rightarrow$

$$\frac{\partial u}{\partial t} = D \nabla \cdot [\nabla u - 2u \nabla \chi_w(u)] + f(u).$$

However: Individuals may migrate over a **network-like structure**.

Nonlocal Directed Motion



Nonlocal Directed Motion

We use the Swing equations as an good example.

Discrete (size N): for $i = 1, \dots, N$,

$$\ddot{\theta}_i(t) + \alpha_i \dot{\theta}_i(t) = P_i + \frac{1}{N} \sum_{j=1}^N K_{ij} \sin(\theta_j(t) - \theta_i(t)). \quad (4.1)$$

Step-function extensions on $[0, 1]$:

Let $I_i := (\frac{i-1}{N}, \frac{i}{N}]$. Define

$$\begin{aligned} \theta^{(N)}(t, x) &:= \theta_i(t) \text{ for } x \in I_i, & \alpha^{(N)}(x) &:= \alpha_i, & P^{(N)}(x) &:= P_i, \\ K^{(N)}(x, y) &:= K_{ij} \text{ for } (x, y) \in I_i \times I_j. \end{aligned}$$

Continuum form:

$$\begin{aligned}\ddot{\theta}^{(N)}(t, x) = & -\alpha^{(N)}(x) \dot{\theta}^{(N)}(t, x) + P^{(N)}(x) \\ & + \int_0^1 K^{(N)}(x, y) \sin(\theta^{(N)}(t, y) - \theta^{(N)}(t, x)) \, dy.\end{aligned}$$

For $x \in I_i$,

$$\begin{aligned} & \frac{1}{N} \sum_{j=1}^N K_{ij} \sin(\theta_j - \theta_i) \\ &= \sum_{j=1}^N \int_{I_j} K_{ij} \sin(\theta^{(N)}(t, y) - \theta^{(N)}(t, x)) \, dy \\ &= \int_0^1 K^{(N)}(x, y) \sin(\theta^{(N)}(t, y) - \theta^{(N)}(t, x)) \, dy. \end{aligned}$$

Graphon limit: if $K^{(N)} \rightarrow K$ (a.e. in L^2) and $\theta^{(N)} \rightarrow \theta$,

$$\partial_{tt}\theta + \alpha(x)\partial_{tt}\theta = P(x) + \int_0^1 K(x, y) \sin(\theta(t, y) - \theta(t, x)) \, dy.$$

Summary

- Continuum limits connect discrete systems to PDE frameworks.
- Enable analytical tools (stability, bifurcation) for large networks.

Thank you!